Testing Evidence Summary

Final report testing chapter - verified outcomes and remaining execution evidence

Scope and provenance

This summary follows SENG 34213 System Development Project Guideline sections 6.2, 6.4, 10.1, and 10.4. It distinguishes local test results, CI evidence, source-level endpoint links, and live environment results.

Test repository branch: feature/TEST-042-final-testing-deliverables; summary source SHA: a1ed48b4ff95456e198433df464985a0caea65d2. The repository register contains 663 requirement-linked case rows. The route matrix contains 235 effective HTTP controller route rows.

Test repository CI: run 40182778270; source SHA a1ed48b4ff95456e198433df464985a0caea65d2; conclusion success.

Test execution results

The test repository Maven clean verify completed successfully: 103 JUnit methods passed. The evidence register maps those results to 117 Passed case rows; 546 rows are Not Run, with zero Failed, Skipped, or Unmapped rows. Multiple case rows may map to one JUnit method.

The 235-row route matrix records source-linked request/assertion categories: 162 success and negative assertions, 46 success assertions, and 27 negative assertions only. These are implementation links; they do not prove live integration execution for every route.

Historical live evidence is incomplete: the documented run had 17 methods pass and 23 mapped rows pass, while 640 rows were not run. That run did not retain a test SHA or GitHub run ID, so it is not used as current release evidence.

Frontend: 5 Student/Admin Playwright E2E tests passed locally. The AI chat E2E was excluded under the team's decision to verify AI manually. The tested E2E files match the frontend development branch; no frontend code change was needed.

Coverage results and guideline interpretation

The guideline's 80% line and branch target applies to new or modified production code. Backend TEST-042 added tests, JaCoCo configuration, CI, and documentation; it did not change production Java source. Therefore the whole-backend aggregate is not the 80% acceptance gate for this branch.

Latest available backend-wide aggregate baseline: run 40172276922, SHA 40deaa168180365c3f9370aefcffe92471c58c26. It measured 12 unit modules at 53.28% line, 33.73% branch, and 50.55% instruction. This is historical context only, not a current-run aggregate or the changed-production-code result for TEST-042.

Current backend unit/coverage CI: run 40180383416; source SHA 8cb633c2c12868cb3239a8940e24cad1abd42746; conclusion success. All 14 unit/coverage matrix jobs passed and 12 JaCoCo module artifacts were published, including assignment-service.

SDS critical logic: the Maven build enforces at least 90% branch coverage independently for QuizGradingServiceImpl (SDS quiz auto-grading, section 4.5) and GradingServiceImpl (SDS assignment grading, section 4.6). Both measured 100% line and branch coverage locally: Quiz 35/35 lines and 14/14 branches; Assignment 68/68 lines and 38/38 branches. The full assignment-service verify passed 95 tests, and its backend CI job passed with a JaCoCo artifact.

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Guideline acceptance matrix

Criterion	Guideline target	Evidence	Assessment
New/changed production code	>=80% line and branch	TEST-042 changes no backend production Java source; this branch does not claim a whole-backend 80% aggregate.	Target scope: not triggered by this branch.
API integration	At least one passing integration case per effective endpoint	235 route rows have source links; 546 register rows remain Not Run; no current live endpoint run is recorded.	Incomplete; run the PR-triggered backend integration workflow.
SDS critical grading	>=90% branch per scoped class	QuizGradingServiceImpl: 100% (14/14); GradingServiceImpl: 100% (38/38); backend CI verify passed.	Met for the two documented grading classes.
Performance	Measured bounded run and reported latency/throughput/errors	No performance measurements or successful workflow run are recorded.	Pending dedicated-performance run.
Staging smoke	Approved read-only staging run with run ID and deployed SHA	No staging run ID, deployment SHA, or smoke artifact is recorded.	Pending staging run.
AI verification	Team's chosen manual verification	AI chat E2E was excluded; the team reports manual verification.	No automated/live AI result is claimed.

Not Run/Pending is a recorded execution state, not a failed test result. Update the relevant report only after a successful run produces its JUnit/JaCoCo or sanitized environment artifact.